

Amazon Dataset Analysis Report

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Executive Summary This report provides a comprehensive analysis of the Amazon product dataset containing 1,465 product listings across 211 unique categories. The analysis covers pricing patterns, customer ratings, discount trends, and category performance.

Dataset Overview - **Total Products**: 1,465 - **Unique Products**: 1,351 - **Unique Categories**: 211 - **Unique Users**: 1,194 - **Unique Reviews**: 1,194

Key Findings

1. Pricing Analysis - **Average Discounted Price**: \$3,126 - **Average Actual Price**: \$5,447 - **Average Discount Percentage**: 47.7% - **Average Customer Rating**: 4.1/5.0

2. Price Range Distribution | Price Range | Product Count | Average Rating |
|-----|-----|-----| | Under \$500 | 576 | 4.07 | | \$500-\$2000 | 527 | 4.09 | | \$2000-\$10000 | 239 | 4.11 | | \$10000-\$50000 | 119 | 4.19 | | Over \$50000 | 3 | 4.37 |

Insight: Higher-priced products tend to have slightly better ratings, suggesting a correlation between price and quality/customer satisfaction.

3. Top Product Categories by Volume | Category | Product Count | Avg Rating | Avg Discount |
|-----|-----|-----|-----| | USB Cables | 233 | 4.15 | 59.4% | | Smart Watches | 76 | 4.03 | 69.8% | | Smartphones | 68 | 4.10 | 23.3% | | Smart TVs | 63 | 4.21 | 38.3% | | In-Ear Headphones | 52 | 3.90 | 58.3% |

Insight: USB cables dominate the dataset with 233 products, likely due to the variety of types and brands available. Smart watches show the highest average discount at 69.8%.

4. Most Reviewed Products | Product | Rating | Review Count | Discount | Price |
 |-----|-----|-----|-----|-----| | AmazonBasics HDMI Cable (3-Foot) | 4.4 | 426,973 | 69% | ■219 | | AmazonBasics HDMI Cable (6-Foot) | 4.4 | 426,973 | 35% | ■309 | | boAt Bassheads 100 Earphones | 4.1 | 363,713 | 65% | ■349 | | Redmi 9 Activ Smartphone | 4.1 | 313,836 | 23% | ■8,499 | | Redmi 9A Sport Smartphone | 4.1 | 313,836 | 24% | ■6,499 |

****Insight**:** AmazonBasics products dominate the most-reviewed list, indicating strong brand trust and volume sales. Budget-friendly electronics like earphones and smartphones also show high engagement.

5. Discount Analysis - ****Highest Discount Categories**:** Smart Watches (69.8%), USB Cables (59.4%), Remote Controls (59.5%) - ****Lowest Discount Categories**:** Smartphones (23.3%), Smart TVs (38.3%) - ****Overall Discount Trend**:** Most products offer significant discounts (47.7% average), suggesting competitive pricing strategies

Strategic Recommendations

For Sellers: 1. ****Focus on Quality**:** Higher-priced products with better ratings suggest customers value quality over low prices 2. ****Competitive Pricing**:** The high average discount (47.7%) indicates price sensitivity among customers 3. ****Category Expansion**:** USB cables and accessories show high volume, suggesting opportunities in accessory markets

For Buyers: 1. ****Value for Money**:** Products in the ■500-■2000 range offer good balance of price and quality 2. ****Brand Reliability**:** AmazonBasics products show consistent quality and high customer satisfaction 3. ****Discount Timing**:** Smart watches and accessories frequently offer high discounts (60-70%)

For Market Analysis: 1. ****Electronics Dominance**:** The dataset is heavily skewed toward electronics and accessories 2. ****Customer Engagement**:** High review counts indicate active customer participation 3. ****Price-Quality Relationship**:** Positive correlation between price and ratings suggests market efficiency

Conclusion The Amazon dataset reveals a marketplace focused on electronics and accessories, with competitive pricing driving customer engagement. The strong performance of AmazonBasics products and the high volume of budget-friendly

electronics indicate a value-conscious customer base. Sellers should focus on maintaining quality while offering competitive discounts to succeed in this marketplace.

--- *Report generated on 2025-06-18* *Data Source: Amazon Product Dataset*